

CYPRESS HILLS PARK, SK, Canada

WMO: 711390

Lat: 49.6406N

Lon: 109.5144W

Elev: 1271

StdP: 86.96

Time Zone: -6.00 (W06)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) -28.7	(c) -25.8	(d) -32.2	(e) 0.2	(f) -28.4	(g) -29.1	(h) 0.3	(i) -25.7	(j) 10.5	(k) 1.8	(l) 9.5	(m) 0.7	(n) 3.3	(o) 330	(p) 0.658

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a) 7	(b) 11.3	(c) 28.4	(d) 14.9	(e) 26.6	(f) 14.6	(g) 24.6	(h) 14.2	(i) 17.1	(j) 24.0	(k) 16.1	(l) 23.3	(m) 15.1	(n) 22.4	(o) 3.9	(p) 180

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
14.8	12.2	19.7	13.4	11.2	18.7	12.2	10.4	17.6	53.1	23.9	49.5	23.5	46.6	22.4	22.3

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a) 8.9	(b) 7.9	(c) 7.1	(e) -31.4	(f) 31.4	(g) 3.6	(h) 1.8	(i) -34.1	(j) 32.7	(k) -36.2	(l) 33.7	(m) -38.2	(n) 34.7	(o) -40.9	(p) 36.0		
(a) 8.9	(b) 7.9	(c) 7.1	(e) -31.6	(f) 19.2	(g) 3.5	(h) 1.5	(i) -34.1	(j) 20.3	(k) -36.2	(l) 21.1	(m) -38.2	(n) 22.0	(o) -40.7	(p) 23.1		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	3.9	-8.1	-7.7	-3.2	3.1	8.7	12.9	17.1	16.6	11.5	4.3	-2.5	-7.4
	DBStd	11.07	8.89	8.06	7.56	5.63	4.83	3.82	3.57	4.10	5.40	5.73	7.03	7.98
	HDD10.0	2930	561	496	408	214	82	14	1	3	44	190	377	539
	HDD18.3	5357	819	729	666	457	299	167	67	81	213	436	626	798
	CDD10.0	686	0	0	0	7	42	102	222	209	89	13	1	0
	CDD18.3	71	0	0	0	0	1	4	29	29	8	0	0	0
	CDH23.3	686	0	0	0	1	11	43	273	282	75	1	0	0
	CDH26.7	149	0	0	0	0	1	6	61	71	12	0	0	0
(14) Wind	WSAvg	3.5	3.7	3.5	3.8	3.7	3.7	3.3	3.1	3.0	3.2	3.7	3.6	3.7
(15) Precipitation	PrecAvg	449	25	23	30	41	50	82	52	43	35	26	20	21
	PrecMax	717	74	42	58	90	129	262	97	132	72	127	47	46
	PrecMin	221	5	3	8	11	1	22	9	0	0	0	3	0
	PrecStd	107	17	12	14	20	38	58	30	34	23	27	12	12
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	10.4	9.8	13.7	20.4	25.0	27.4	30.5	30.9	28.4	21.1	14.3	9.8
		MCWB	3.7	3.5	6.3	9.4	12.3	15.5	16.0	15.2	13.7	11.0	6.7	2.9
	2%	DB	7.0	6.7	10.5	17.1	22.1	24.2	28.4	28.7	25.7	17.6	11.1	6.6
		MCWB	2.2	2.1	4.4	8.1	11.5	14.8	15.7	14.4	13.0	9.1	5.0	2.0
	5%	DB	5.0	4.4	8.2	14.3	19.4	21.9	26.5	26.6	22.9	14.7	8.7	4.7
		MCWB	1.1	0.6	3.2	6.5	10.3	13.7	15.4	14.3	12.2	7.7	3.7	1.0
	10%	DB	3.2	2.4	6.2	11.5	17.0	20.0	24.4	24.4	20.0	12.3	6.5	2.6
		MCWB	0.0	-0.7	1.9	4.9	9.0	12.5	15.0	13.9	10.9	6.3	2.3	-0.5
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	4.8	4.2	6.7	10.2	13.7	17.6	19.2	17.7	15.4	11.8	7.6	4.0
		MCDB	9.1	8.3	12.7	18.8	21.4	23.1	25.9	24.4	24.3	19.6	13.7	7.9
	2%	WB	2.6	2.4	5.0	8.5	12.1	15.8	17.7	16.5	13.9	9.6	5.4	2.3
		MCDB	6.1	6.2	9.5	16.0	20.3	22.0	24.7	23.9	23.0	16.1	10.2	6.0
	5%	WB	1.3	0.9	3.5	7.0	10.9	14.4	16.5	15.5	12.6	8.1	4.0	1.0
		MCDB	4.8	4.2	7.7	13.3	18.3	20.7	23.7	23.4	21.3	13.7	8.3	4.3
	10%	WB	0.0	-0.6	2.1	5.4	9.6	13.1	15.5	14.6	11.4	6.7	2.4	-0.5
		MCDB	3.1	2.2	6.0	11.0	15.7	19.0	22.6	22.5	19.2	11.7	6.1	2.6
(35) Mean Daily Temperature Range	5% DB	MDBR	8.9	8.4	8.3	9.6	10.2	9.8	11.3	11.4	10.6	9.2	8.0	7.9
		MCDBR	9.2	9.8	10.8	13.5	13.6	12.4	13.8	14.2	13.9	12.3	10.2	9.1
		MCWBR	6.3	6.7	6.6	7.4	6.8	6.7	6.1	5.8	6.4	6.5	6.4	6.2
		MCDBR	9.8	9.9	10.5	12.7	12.4	11.6	12.1	12.2	12.9	11.7	10.0	9.1
	5% WB	MCWBR	7.2	7.2	6.7	7.3	6.6	6.7	6.3	6.0	6.4	6.4	6.7	6.5
(40) Clear-Sky Solar Irradiance	taub		0.248	0.252	0.277	0.319	0.340	0.330	0.335	0.350	0.309	0.278	0.251	0.240
	taud		2.410	2.454	2.478	2.381	2.362	2.448	2.442	2.374	2.465	2.538	2.508	2.411
	Ebn at Noon		829	919	946	929	917	927	917	885	895	874	828	789
	Edh at Noon		61	75	88	109	116	108	107	109	89	69	55	53
(44) All-Sky Solar Radiation	RadAvg		1.31	2.33	3.65	4.90	5.83	6.30	6.86	5.60	4.04	2.52	1.44	1.01
	RadStd		0.12	0.17	0.23	0.33	0.40	0.40	0.42	0.40	0.34	0.25	0.12	0.08

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	-0.76	N/A	N/A	N/A	N/A
(47) Regional (28 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page